

Glossary and Data Dictionary

Document status

Field	Value
Lifecycle position	Stage 2A — Supporting BRD artifact
Document status	Draft v0.1 — Pre-baseline
Validation status	Pending business and Data Owner/Steward validation
Data basis	Current synthetic portfolio data and repository configuration
Repository evidence	HEAD 8026f57 — Gen2 close-out
Dataset configuration	Version 1.0.0 ; schema version 1.0
Snapshot clock	2024-12-30
Approval status	No approval recorded

 This document describes the current synthetic implementation and proposed business terminology. It is not a production schema, legal data classification, data-processing approval or confirmation that the same field meanings apply to a provider or TPA source system.

1. Purpose and scope

This document:

- Establishes common P5 business and technical terminology.
- Describes all physical fields in the three current synthetic datasets.
- Defines the governed semantic metrics currently exposed by P5.
- Records lineage and version metadata.
- Identifies definitions and classifications requiring validation.
- Supports the BRD, SAD, SRS, RTM, interfaces, user stories and UAT.

- Does not replace source-system specifications, mapping specifications, a production data model, DPIA or retention schedule.

2. Glossary

Term	Controlled definition	Status/validation note
P5	A governed Agentic RCM intelligence layer serving executive, analyst and audit/compliance use cases.	Intended product definition; pre-baseline.
Revenue Cycle Management (RCM)	Operational and financial activities supporting claim preparation, submission, payment, denial management, accounts receivable and recovery.	Business interpretation requires stakeholder validation.
Provider	The healthcare-delivery organisation sponsoring or using P5.	Exact organisation and legal role not assigned.
Third-Party Administrator (TPA)	A partner performing delegated payer, claim, denial or appeal activities for or with the provider.	Exact responsibilities and authority remain unresolved.
Payer	The insurer or payer organisation associated with a claim or cohort record.	Current values are synthetic configuration values.
Claim	A record representing a request for payment for healthcare services.	Exact grain and source-system

		meaning require production mapping.
Denial	A claim whose governed status is Denied .	Implemented synthetic status definition.
Denial reason	A controlled category describing why a claim was denied.	Current categories are synthetic and require contract/regulatory validation.
Recoverable denial	A denied claim considered eligible and inside the configured filing window, ranked using governed inputs.	Implemented backend definition; business eligibility and window require validation.
Filing/submission window	The permitted time between service/submission events and an appeal or resubmission.	Current 90-day rule is regulatory/contractual truth outside dataset configuration and requires stakeholder validation.
Overturn probability	The governed historical probability associated with a denial-reason category for ranking purposes.	Current values are synthetic/code evidence; production source and ownership require validation.
Expected recoverable value	Claim amount multiplied by the governed overturn probability for its denial reason.	Computed in tested code; not by the LLM.
Revenue leakage	In the current synthetic RCM dataset, ClaimAmount - AmountPaid .	Demonstration definition only; production business

		definition requires approval.
Clean claim	In the current synthetic generator, an approved claim submitted within seven days.	Demonstration rule only; production definition requires approval.
Clean claim rate	Clean-claim rows divided by total claims, expressed as a percentage.	Computed by the governed semantic view.
AR ageing	Grouping of claims into outstanding-age buckets or Paid .	Current synthetic buckets use submission date and a fixed reference date; production rule requires approval.
Executive Insight	Aggregate-only executive surface for KPIs, governed narratives, charts and sources.	Designed/not implemented.
Analyst Workbench	Analyst surface for questions, denial prioritisation, evidence and appeal preparation.	Designed/not implemented.
Audit Console	Administrative/compliance surface for audit events, evidence and versions.	Designed/not implemented.
Executive/Viewer	Role limited to approved aggregate metrics and viewer-safe functions.	Backend boundary implemented; product authentication/UI pending.

Analyst	Role permitted to investigate approved claim/member detail and prepare decision-support drafts.	Authority does not automatically include appeal approval.
Admin/Auditor	Role permitted to inspect administrative, audit and version evidence.	Final production permissions require approval.
Agent	The LLM orchestration component that selects tools and narrates governed results.	It must not compute or invent numbers.
Tool	A typed, registered operation invoked by the agent through governed dispatch.	Implemented backend concept.
Semantic view	A predefined, parameterised calculation over an approved source; no free-text SQL.	Implemented.
Deterministic rules engine	Versioned code producing compliance verdicts from approved rules.	Implemented backend; production rules require approval.
Grounding validator	Fail-closed control checking numeric statements against trusted tool results or approved constants.	Implemented.
Evidence lineage	Metadata connecting an output to its source, query, parameters,	Partly implemented end to end.

	execution time and applicable versions.	
Query ID	Unique identifier for a governed query result used for citation and chart binding.	Implemented technical identifier.
Result envelope	Standard tool response containing rows/results plus source, parameters, identifiers, time and version metadata.	Implemented pattern.
Dataset generation	An accepted or candidate appended data drop identified by a generation tag.	Implemented Gen2 concept.
Configuration version	Semantic version identifying the dataset configuration used by a result.	Implemented.
Configuration hash	Content hash identifying the precise configuration content used.	Implemented.
as_of	Pinned snapshot date used instead of an uncontrolled real-time clock.	Advances only through accepted onboarding.
Rules version	Version of the deterministic compliance rule set used for a verdict.	Implemented.

Model version	Version of the registered predictive model or metadata used for a score/result.	Metadata exists; production serving is incomplete.
Method version	Version of an analytical/statistical method.	Future Gen3 concept; designed/not implemented.
Golden set	Fixed evaluation questions with governed expected results or behavioural conditions.	Implemented; currently 55 items.
Dry-run evaluation	Validation of the harness, tools and stored truths without live model execution.	Implemented.
Live evaluation	Evaluation that executes the configured model and records transcripts/results.	Release evidence remains incomplete for the documented open gap.
Human in the loop (HITL)	Requirement that a named authorised human performs approval or external action.	Safety principle established; full workflow not implemented.
pending_review	Mandatory status returned by an appeal draft before human action.	Implemented tool contract; full workflow incomplete.

Requirements baseline	Formally approved requirements, scope, priorities, status and governance basis.	Not established.
UAT	Business acceptance testing against approved scenarios and expected results.	Planned; not executed.
Implemented	Present in current source and supported by tests or run evidence.	Controlled delivery status.
Designed/not implemented	Defined by design, architecture or requirements but absent from current source.	Controlled delivery status.
Deferred	Explicitly excluded from the current baseline or assigned to a future phase.	Controlled delivery status.
Assumption requiring validation	Proposed business, operational, security or compliance position needing evidence.	Validation label, not delivery status.

3. Proposed data-handling categories

Status: proposed handling categories only — not a legal classification.

Code	Category	Intended handling
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C0	Technical and lineage metadata	Used for traceability, diagnostics and audit; access remains role-controlled.
C1	Aggregate or controlled business dimension	May support viewer-safe aggregate presentation; raw rows are not automatically viewer-accessible.
C2	Pseudonymous operational record detail	Analyst/admin access only for an approved purpose.
C3	Potentially sensitive personal, clinical or financial data	No viewer-level raw access; production use requires privacy, minimisation, lawful-purpose, retention and access approval.

 All current records are synthetic. The category codes describe intended handling, not confirmed UAE legal classification. Production classification must be approved by the Information Security and Privacy Owner and Compliance and Regulatory Owner.

4. Dataset catalogue

Logical source	Physical dataset/table	Grain	Current rows	Coverage	Governed purpose
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claims_analysis	claims_clean.csv / claims_analysis	One synthetic claim record	4,500	2022-07 to 2024-07	Claim volume, denial rate/value, specialty spend and monthly trends
rcm_ops	rcm_dashboard.csv / rcm_ops	One synthetic operational claim record	3,000	2023-01 to 2024-12	Payer denials, leakage, AR ageing, payment time and clean-claim metrics
cohort	chronic_disease_cohort.csv / cohort	One synthetic chronic-disease member-year record	3,000	2023–2024	Cohort cost, risk-tier, high-cost and model-registry metrics

- Sources must not be blended into one figure.
- A metric is available only when supported by the selected source and current configuration.
- Row counts and coverage describe the current synthetic configuration and may change through an accepted dataset generation.

5. Physical field dictionary

5.1 claims_analysis — all 19 fields

Physical field	Type/format	Definition	Category and governance note
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ClaimID	String/UUID	Synthetic unique claim identifier.	C2; record-level access only.
PatientID	String/UUID	Synthetic patient/member identifier associated with the claim.	C3; not required by approved aggregate metrics; necessity requires validation.
ProviderID	String/UUID	Synthetic provider identifier.	C2; production master-data mapping required.
ClaimAmount	Decimal AED	Amount associated with the claim.	C3; governed measure, but exact billed/allowed meaning requires validation.
ClaimDate	ISO date	Date stored for the synthetic claim.	C2; whether this represents service, submission or another event requires validation.
DiagnosisCode	String	Synthetic diagnosis-code value.	C3; not approved as valid ICD-10 or clinical truth.
ProcedureCode	String	Synthetic procedure-code value.	C3; code system and validity require confirmation.

PatientAge	Integer years	Synthetic age value.	C3; not used by current governed aggregate metrics.
PatientGender	Enum F , M	Synthetic gender value.	C3; limited synthetic categories are not a production standard.
ProviderSpecialty	Controlled string	Provider specialty dimension.	C1 when aggregated; current values are configuration-controlled.
ClaimStatus	Enum Approved , Denied , Pending	Governed claim-status category.	C2 at record level; C1 in aggregate.
PatientIncome	Decimal	Synthetic patient-income value.	C3; no approved P5 business use; necessity should be challenged.
PatientMaritalStatus	Controlled string	Synthetic marital-status value.	C3; no approved P5 business use.
PatientEmploymentStatus	Controlled string	Synthetic employment-status value.	C3; no approved P5 business use.
ProviderLocation	String	Fictional provider-location label.	C1/C2; values are illustrative, not production geography.

ClaimType	Controlled string	Claim-type dimension.	C1 when aggregated.
ClaimSubmissionMethod	Enum Online , Paper , Phone	Synthetic submission-channel value.	C2; not currently a governed metric input.
ClaimYear	Integer YYYY	Year derived for the claim.	C1.
ClaimMonth	String YYYY-MM	Month derived for the claim.	C1.

Current controlled dimensions:

- Claim types: Emergency, Inpatient, Outpatient, Routine.
- Specialties: Cardiology, General Practice, Neurology, Orthopedics, Pediatrics.
- Statuses: Approved, Denied, Pending.

5.2 rcm_ops — all 20 fields

Physical field	Type/format	Definition	Category and governance note
ClaimID	String UAE-NNNNN	Synthetic operational claim identifier.	C2; analyst/admin record-level access.
DateOfService	ISO date	Synthetic service date and current source clock column.	C2; drives accepted as_of .
SubmissionDate	ISO date	Synthetic date the claim was submitted.	C2.
PaymentDate	Nullable ISO date	Synthetic payment date; populated for	C2.

		approved/paid claims.	
Payer	Controlled string	Payer associated with the claim.	C1 when aggregated; source/master-data mapping required.
ClaimType	Controlled string	Operational claim-type dimension.	C1 when aggregated.
Specialty	Controlled string	Operational specialty dimension.	C1 when aggregated.
Facility	Controlled string	Synthetic healthcare-facility label.	C1 when aggregated; values are illustrative.
ClaimAmount	Decimal AED	Synthetic amount associated with the claim.	C3; exact financial meaning requires approval.
AmountPaid	Decimal AED	Synthetic amount paid; zero for non-approved records in the current generator.	C3; production treatment of zero versus null requires approval.
RevenueLeakage	Decimal AED	Current synthetic calculation: $\text{ClaimAmount} - \text{AmountPaid}$	C3; proposed demonstration definition only.

ClaimStatus	Enum Approved , Denied , Pending	Operational claim status.	C2 at record level; C1 in aggregate.
DenialReason	Nullable controlled string	Denial category; populated for denied claims in the synthetic source.	C2; categories require production validation.
DaysToPayment	Nullable number of days	Days from submission to payment for paid claims.	C2; used by the governed average- payment metric.
SubmissionLag	Integer days	Days between service and submission in the synthetic source.	C2.
CleanClaim	Binary 0/1	Current synthetic flag: 1 when approved and submission lag is seven days or less.	C2; production definition requires approval.
ARAgeing	Controlled bucket	Current bucket: 0-30 , 31-60 , 61-90 , 90+ or Paid .	C2; synthetic reference-date logic requires replacement/app roval.
ServiceYear	Integer YYYY	Year derived from service date.	C1.
ServiceMonth	String YYYY-MM	Month derived from service	C1.

		date.	
ServiceQ	String YYYYQn	Quarter derived from service date.	C1.

Current controlled values:

- Payers: Daman (HAAD), ADNIC, AXA Gulf, BUPA Global, Cigna, MetLife, Sukoon Insurance.
- Claim types: Inpatient, Outpatient, Emergency, Daycase, Dental, Maternity.
- Specialties: Pediatrics, Cardiology, Orthopedics, Internal Medicine, Obstetrics, Oncology, General Surgery, Dermatology.
- Denial reasons: Missing Pre-authorisation; Coding Error (ICD-10); Duplicate Claim; Patient Not Eligible; Exceeded Policy Limit; Late Submission; Missing Documentation; Service Not Covered.
- Statuses: Approved, Denied, Pending.
- AR buckets: 0-30 days; 31-60 days; 61-90 days; 90+ days; Paid.

5.3 cohort — all 25 fields

Physical field	Type/format	Definition	Category and governance note
member_id	String MBR-NNNNN	Synthetic pseudonymous member identifier.	C3; analyst/admin only.
year	Integer YYYY	Synthetic member-year period.	C1 when aggregated.
emirate	Controlled string	Synthetic emirate category.	C3 at record level; C1 when aggregated.
nationality	Controlled string	Synthetic nationality grouping.	C3; no approved current semantic metric; necessity

			requires validation.
gender	Controlled string	Synthetic gender value.	C3.
age	Integer years	Synthetic member age.	C3.
plan_tier	Enum Basic , Enhanced , Premium	Synthetic insurance-plan tier.	C3; production meaning requires payer mapping.
payer	Controlled string	Synthetic payer category.	C3 at member level; C1 when aggregated.
condition	Controlled string	Synthetic chronic-condition grouping.	C3; not a clinical determination by P5.
bmi	Decimal	Synthetic body-mass-index value.	C3 clinical data.
hba1c	Nullable decimal	Synthetic HbA1c value; null where not generated/applicable.	C3 clinical data.
systolic_bp	Integer	Synthetic systolic blood-pressure value.	C3 clinical data.
smoker	Binary 0/1	Synthetic smoking indicator.	C3 clinical/personal data.

comorbidities	Integer count	Synthetic number of comorbidities.	C3 clinical data.
medication_compliance	Enum High, Medium, Low	Synthetic medication-compliance category.	C3; not a validated clinical assessment.
er_visits	Integer count	Synthetic emergency-room visit count.	C3 utilisation data.
inpatient_days	Integer count	Synthetic inpatient-day count.	C3 utilisation data.
outpatient_visits	Integer count	Synthetic outpatient-visit count.	C3 utilisation data.
specialist_visits	Integer count	Synthetic specialist-visit count.	C3 utilisation data.
total_cost_paid	Integer AED	Synthetic annual total member cost.	C3 financial data.
pharmacy_cost_aed	Integer AED	Synthetic pharmacy-cost component.	C3 financial/clinical data.
procedure_cost_aed	Integer AED	Synthetic procedure-cost component.	C3 financial/clinical data.
consultation_cost_aed	Integer AED	Synthetic consultation-cost component.	C3 financial/clinical data.

<code>risk_tier</code>	Enum <code>Low</code> , <code>Medium</code> , <code>High</code>	Synthetic tier based on annual-cost percentiles: approximately 33rd and 66th percentile boundaries.	C3; portfolio segmentation only, not a production clinical or coverage classification.
<code>high_cost_flag</code>	Binary <code>0/1</code>	Synthetic flag for annual cost at or above the 80th percentile.	C3; must not drive coverage, claim or premium decisions.

Current controlled dimensions:

- Emirates: Dubai, Abu Dhabi, Sharjah, Ajman, Other.
- Conditions: Diabetes; Hypertension; Diabetes + Hypertension; Cardiac; Diabetes + Cardiac.
- Payers: Daman (HAAD), ADNIC, AXA Gulf, BUPA Global, Cigna, MetLife, Sukoon Insurance.
- Risk tiers: Low, Medium, High.
- High-cost flag: 0 = not flagged; 1 = flagged.

6. Governed semantic metric dictionary

Metric	Valid source(s)	Controlled calculation/output	Access/governance note
<code>denial_rate</code>	<code>claims_analyses</code> , <code>rcm_ops</code>	Denied claims divided by total claims; returns counts and <code>denial_rate_pct</code> .	Computed server-side; no LLM division.
<code>claims_volume</code>	<code>claims_analyses</code> , <code>rcm_ops</code>	Claim count, optionally grouped/filtered. Ungrouped	Aggregate viewer-safe; record-level <code>top_n</code>

		top_n returns ranked claim detail.	analyst/admin only.
denied_value	claims_analysis, rcm_ops	Governed aggregation of claim amount where status is Denied .	Exact financial interpretation depends on approved ClaimAmount meaning.
spend_by_specialty	claims_analysis	Sum/count/average or approved aggregation of claim amount by specialty or supported dimension.	Synthetic demonstration measure.
monthly_trend	claims_analysis	Monthly claim count, denied count, denial rate and total claim amount.	Server-computed.
leakage_by_reason	rcm_ops	Sum of current RevenueLeakage ; grouped results include server-computed shares and totals.	Includes underpayment on approved claims; production leakage definition requires approval.
ar_ageing	rcm_ops	Claim count and claim value by configured AR bucket.	Current bucket logic is synthetic.

avg_days_to_payment	rcm_ops	Average DaysToPayment over paid rows; returns rounded headline and exact mean.	Server-computed.
clean_claim_rate	rcm_ops	Rows with CleanClaim = 1 divided by total rows.	Current clean-claim rule is synthetic.
denials_by_payer	rcm_ops	Denied claim count and denied claim value grouped by payer or supported dimension.	Server-computed.
cohort_cost	cohort	Approved aggregation of total_cost_approved , optionally grouped.	Aggregate viewer-safe; member filter/detail restricted.
risk_tiers	cohort	Member count, total cost and average cost by risk tier.	Synthetic cost-derived segmentation.
high_cost_flags	cohort	Flagged-member count, total members and server-computed flagged share.	No coverage, claims or premium use.

<code>model_scores</code>	<code>cohort</code>	Aggregate model-registry metadata and held-out model metrics.	Viewer-safe metadata; not live member scoring.
<code>claim_lookup</code>	<code>claims_analys</code> <code>sis</code> , <code>rcm_ops</code>	Complete row for one governed claim ID.	Analyst/admin only; viewer prohibited.

i `score_member`, `check_compliance`, `get_recoverable_denials`, `draft_appeal` and `render_chart` are governed tools rather than dataset metrics. Production `score_member` serving and the complete appeal workflow are not implemented.

7. Technical lineage and identifier dictionary

Field	Meaning
<code>source</code>	Logical governed dataset used.
<code>view_name</code>	Semantic view or tool calculation executed.
<code>params_echo</code>	Validated parameters used by the calculation.
<code>query_id</code>	Identifier for the governed query result and citation.
<code>executed_at</code>	Execution timestamp recorded for the result.
<code>row_count</code>	Number of rows returned by the governed result.
<code>config_version</code>	Dataset-configuration semantic version.

<code>config_hash</code>	Hash of the exact dataset configuration content.
<code>dataset_generation</code>	Accepted appended data generation serving the result.
<code>rules_version</code>	Compliance-rules version used for a verdict.
<code>model_version</code>	Predictive-model version associated with a score or metadata result.
<code>method_version</code>	Future analytical-method version used for a Gen3 result.
<code>check_id</code>	Identifier for a compliance-check execution.
<code>score_id</code>	Identifier for a scoring execution.
<code>draft_id</code>	Identifier for an appeal draft.
<code>chart_id</code>	Identifier for a rendered chart bound to an existing query.

8. Data governance rules

1. `config/datasets.yaml` is the implemented source of truth for dataset-shaped facts.
2. Regulatory rules and the filing window are not dataset configuration.
3. Different governed sources must not be blended into one figure.
4. Unsupported source, metric, dimension, filter or enum combinations must be rejected rather than returning misleading empty results.
5. Raw agent-plane data access remains read-only.
6. Ingestion uses a separate controlled operations path.
7. Accepted extracts append with a dataset-generation tag.
8. `as_of` advances only through human-accepted onboarding.
9. Rejected onboarding restores data, configuration, generated artefacts and truths.
10. Viewer users receive aggregates only.
11. Claim/member detail requires an approved analyst/admin purpose.
12. UI mock-up figures and identities are illustrative and never test truth.

13. Synthetic data does not establish production privacy, security, quality or legal suitability.
14. Unknown or unmapped production fields must remain unknown until reviewed; meanings must never be silently inferred in the Gen2 baseline.

9. Open data-definition decisions

ID	Open decision	Proposed owner role	Required by
DD-01	Confirm the provider and TPA systems of record and data-controller/processor responsibilities.	Product Owner with Compliance and Regulatory Owner	Requirements baseline
DD-02	Approve the exact business meaning of claim amount, paid amount and leakage for each source.	Data Owner/Steward with Provider RCM/AR Operations Owner	Requirements baseline
DD-03	Confirm the event represented by <code>claims_analys</code> <code>sis.ClaimDate</code> .	Data Owner/Steward	SRS/interface design
DD-04	Approve production clean-claim rules and exclusions.	Provider RCM/AR Operations Owner	Requirements baseline
DD-05	Approve AR ageing anchor dates, bucket	Provider RCM/AR	Requirements baseline

	rules and treatment of pending/denied claims.	Operations Owner	
DD-06	Approve filing-window and overturn-rate sources, ownership and versioning.	Compliance and Regulatory Owner with Provider/TPA operations	Requirements baseline
DD-07	Confirm source grain, primary keys, duplicate rules and null handling.	Data Owner/Steward	SRS/interface design
DD-08	Approve production enumerations, aliases and code mappings.	Data Owner/Steward	SRS/interface design
DD-09	Decide whether income, marital status, employment status, nationality and clinical attributes are necessary for P5.	Information Security and Privacy Owner with Product Owner	Requirements baseline
DD-10	Approve production data classifications, purposes, masking,	Information Security and Privacy Owner	Requirements baseline

	retention and role access.		
DD-11	Approve pseudonymisation and authorised re-identification arrangements.	Information Security and Privacy Owner	UAT entry
DD-12	Approve data-quality thresholds, reconciliation controls and generation-acceptance authority.	Data Owner/Steward	UAT entry
DD-13	Approve whether synthetic risk tiers/high-cost flags have any production analogue.	Product Owner with Compliance/Data owners	SRS scope
DD-14	Confirm source-to-target mappings for the first representative provider/TPA extract.	Data Owner/Steward	UAT entry

These are open questions, not approved decisions. They should be added or cross-referenced in the RAID/Decision Log during the later BRD alignment pass.

10. Related artifacts

- [Vision and Strategy](#)
- [Business Requirements Document](#)

- [Stakeholder Register and RACI](#)
- [As-Is and To-Be Processes](#)
- [Compliance, Privacy and Security Requirements](#)
- [Requirements Traceability Matrix](#)
- [RAID and Decision Log](#)
- [Non-Functional and Operational Requirements](#)
- [UI Design Inventory and Acceptance Map](#)